SRM University-AP, Amaravati
Department of Mathematics
Discrete Mathematics (CSE 203)
Practice Problems - Unit 1: The Foundations: Logic and Proofs

1. Which of the following sentences are propositions? What are the truth values of those which are propositions?

   (a) Vijayawada is the capital of India.
   (b) $6 + 3 = 5$.
   (c) $3 + 7 = 10$.
   (d) $x + 3 = 9$.
   (e) Give me a pen.
   (f) Who are you?
   (g) The sun rises in the east.
   (h) $1 + x = 2,\ \forall x \neq 1$ and domain is the set of real numbers.
   (i) $\exists x$ such that $2 + x = 3$ and domain is the set of real numbers.
   (j) $3 + x = 5,\ \forall x \neq 1$ and domain is the set of real numbers.
   (k) $n^2 \geq 50$.

2. Let p, q, and r be the propositions

   p : You have fever.
   q : You miss the final examination.
   r : You pass the course.

   Express each of these propositions as an English sentence.

   (a) $p \to q$
   (b) $\neg q \leftrightarrow r$
   (c) $q \to \neg r$
   (d) $p \vee q \vee r$
   (e) $(p \to \neg r) \vee (q \to \neg r)$
   (f) $(p \wedge q) \vee (\neg q \wedge r)$

3. Determine whether each of these conditional statements is true or false.

   (a) If $1 + 1 = 2$, then $2 + 2 = 5$.
   (b) If $1 + 1 = 3$, then $2 + 2 = 4$.
   (c) If $1 + 1 = 3$, then $2 + 2 = 5$.
   (d) If monkeys can fly, then $1 + 1 = 3$.

4. Determine whether these biconditional statements are true or false.

   (a) $2 + 2 = 4$ if and only if $1 + 1 = 2$.

(b) $1 + 1 = 2$ if and only if $2 + 3 = 4$.

(c) $1 + 1 = 3$ if and only if monkeys can fly.

(d) $0 > 1$ if and only if $2 > 1$.

5. State the converse, contrapositive, and inverse of each of these conditional statements.

(a) If it rains, then I will stay at home.

(b) I go to the beach whenever I am free.

(c) When I stay up late, it is necessary that I sleep until noon.

6. Construct the truth table for each of these compound propositions.

(a) $p \wedge \neg p$

(b) $(p \vee \neg q) \rightarrow q$

(c) $(p \vee q) \rightarrow (p \wedge q)$

(d) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$

(e) $(p \rightarrow q) \rightarrow (q \rightarrow p)$

(f) $(p \oplus q) \rightarrow (p \wedge q)$

(g) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$

(h) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$

(i) $(p \rightarrow q) \wedge (\neg p \rightarrow r)$

(j) $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$

(k) $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$

7. Use truth tables to verify these equivalences.

(a) $p \wedge T \equiv p$

(b) $p \wedge F \equiv F$

(c) $p \vee p \equiv p$

(d) $p \vee F \equiv p$

(e) $p \vee T \equiv T$

(f) $p \wedge p \equiv p$

(g) $p \vee q \equiv q \vee p$

(h) $p \wedge q \equiv q \wedge p$

(i) $(p \vee q) \vee r \equiv p \vee (q \vee r)$

(j) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

8. Show that each of these conditional statements is a tautology by using truth tables.

(a) $[\neg p \wedge (p \vee q)] \rightarrow q$

(b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

(c) $[p \wedge (p \rightarrow q)] \rightarrow q$

(d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

9. Show that $\neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent.

10. Show that $(p \to q) \vee (p \to r)$ and $p \to (q \vee r)$ are logically equivalent.

11. Show that $\neg p \to (q \to r)$ and $q \to (p \vee r)$ are logically equivalent.

12. Show that $(p \vee q) \wedge (\neg p \vee r) \to (q \vee r)$ is a tautology.

13. Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.

    (a) $\forall x(C(x) \to F(x))$
    (b) $\exists x(C(x) \to F(x))$
    (c) $\forall x(C(x) \wedge F(x))$
    (d) $\exists x(C(x) \wedge F(x))$

14. Let $Q(x)$ be the statement "$x + 2 > 2x$". If the domain consists of all integers, what are these truth values?

    (a) $Q(0)$
    (b) $Q(-1)$
    (c) $Q(1)$
    (d) $\exists x Q(x)$
    (e) $\forall x Q(x)$
    (f) $\exists x \neg Q(x)$
    (g) $\forall x \neg Q(x)$

15. Determine the truth value of each of these statements if the domain consists of all real numbers.

    (a) $\exists x(x^5 = 5)$
    (b) $\exists x(x^2 < 3x)$
    (c) $\forall x((-x)^4 = x^4)$
    (d) $\forall x(2x > x + 2)$

16. What are the negations of the statement $\forall x(x^4 > x^2)$ and $\exists x(x^3 = 7)$?

17. Let $S(x)$ be the statement "x is a student", $F(x)$ the statement "x is a faculty", and $A(x, y)$ the predicate "x has asked a question to y", where the domain consists of all people associated with your college. Use quantifiers to express each of these statements.

    (a) Every student has asked Professor Gross a question.
    (b) Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.
    (c) Some students has not asked any faculty member a question.
    (d) There is a faculty member who has never been asked a question by a student.
    (e) Some student has asked every faculty member a question.
    (f) There is a faculty member who has asked every other faculty member a question.
    (g) Some student has never been asked a question by a faculty member.

18. Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.

    (a) The product of two negative integers is positive.

    (b) The average of two positive integers is positive.

    (c) The difference of two negative integers is not necessarily negative.

    (d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.

19. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

    (a) $\exists x \forall y(xy = y)$

    (b) $\forall x \forall y(((x < 0) \wedge (y > 0)) \rightarrow (xy < 0))$

    (c) $\forall x \forall y \exists z(x + y = 5z)$

20. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

    (a) $\forall x \exists y(x^2 = 3y)$

    (b) $\exists x \forall y(xy = 0)$

    (c) $\forall x(x \neq 0 \rightarrow \exists y(xy = 1))$

    (d) $\exists x \forall y(y \neq 0 \rightarrow xy = 1)$

    (e) $\forall x \exists y(x + y = 1)$

    (f) $\exists x \exists y(x + 2y = 2 \wedge 2x + 4y = 5)$

    (g) $\forall x \exists y(x + y = 2 \wedge 2x - y = 1)$

    (h) $\forall x \forall y \exists z \left(z = \frac{x+y}{2}\right)$

    (i) $\forall x \exists y(x = y^2)$

    (j) $\exists x \exists y(x + y \neq y + x)$

21. Rewrite each of these statements so that negations appear only within predicates(that is, so that no negation is outside a quantifier or an expression involved logical connectives).

    (a) $\neg \exists y \exists x P(x, y)$

    (b) $\neg \exists y(Q(y) \wedge \forall x \neg R(x, y))$

    (c) $\neg \exists y(\exists x R(x, y) \vee \forall x S(x, y))$

    (d) $\neg \exists y(\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$

    (e) $\neg \forall x \exists y P(x, y)$

22. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

    (a) $\forall x \forall y(x^2 = y^2 \rightarrow x = y)$

    (b) $\forall x \exists y(y^2 = x)$

    (c) $\forall x \forall y(xy > x)$

23. Establish the validity of the following argument forms using the rules of inferences.

(a)

$$\begin{array}{l} p \to q \\ \neg q \\ \neg r \\ \hline \therefore\ \neg(p \vee r) \end{array}$$

(b)

$$\begin{array}{l} p \vee q \\ \neg r \\ \neg p \vee r \\ \hline \therefore\ q \end{array}$$

(c)

$$\begin{array}{l} p \vee q \\ p \to r \\ r \to s \\ \hline \therefore\ q \vee s \end{array}$$

(d)

$$\begin{array}{l} \neg q \\ \neg r \\ \neg p \vee q \vee r \\ \hline \therefore\ \neg p \end{array}$$

(e)

$$\begin{array}{l} (\neg p \vee q) \to r \\ r \to (s \vee t) \\ \neg s \wedge \neg u \\ \neg u \to \neg t \\ \hline \therefore\ p \end{array}$$

(f)

$$\begin{array}{l} p \to q \\ \neg r \vee s \\ p \vee r \\ \hline \therefore\ \neg q \to s \end{array}$$

24. Use a direct proof to show that the product of two odd numbers is odd.

25. Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

26. Prove that if m and n are integers and mn is even,then m is even or n is even.

27. Show that if n is an integer and $n^3 + 5$ is odd,then n is even using a proof by contradiction.

28. Let $P(n)$ be the proposition "If a and b are positive real numbers, then $(a + b)^n \geq a^n + b^n$." Prove that $P(1)$ is true. What kind of proof did you use?

29. Use a proof by contradiction to show that there is no rational number r for which $r^3 + r + 1 = 0$.

30. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.